P-256 factor-base second pass: prepared source and explicit execution gates

26 October 2026. Six sub-agent groups worked across the eight requested tracks. Seven continuation packages are prepared and their protocols frozen prospectively. F1 is a preserved partial implementation after a platform worker halt. No second-pass mathematical panel, pilot or experimental row has run. The first pass remains a separate immutable evidence layer.

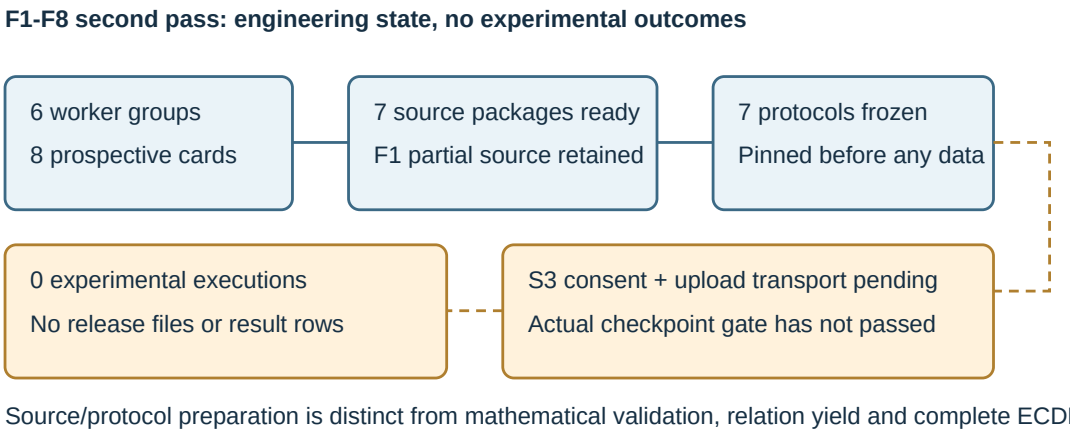

Figure 1: Second-pass state; counts describe engineering preparation only

Track	Prepared continuation	Current state
F1	Quadratic-chain quotient/exterior operators, proposed extraction proof and fixed inputs	Partial; driver, independent oracle and protocol missing after worker halt
F2	Full 65,536-coordinate target-lift classification and 24 signed queries over six matched domains	Source prepared; protocol frozen; unexecuted
F3	Four-partition MITM with exhaustive Jacobian oracle, eight unplanted cases and distinct useful/zero-Q controls	Source prepared; protocol frozen; unexecuted
F4	Sparse weighted incidence reducer, witness DAG and 128 all-draw anchored sections with graph controls	Source prepared; protocol frozen; unexecuted
F5	Matched source/endpoint membership and relation transport through all four certified isogenies	Source prepared; protocol frozen; unexecuted
F6	Root-independent offsets, 60 alternative-shift width cells, ordinary-target extraction and labelled positive control	Source prepared; protocol frozen; unexecuted
F7	Balanced genus-two classes with both infinities, norm and separate extension-field support checks	Source prepared; protocol frozen; unexecuted
F8	Representatives without supplied scalar labels, complete fixed target queries and symmetry-adjusted ranks	Source prepared; protocol frozen; unexecuted

The source snapshot is de1f9bbe50801ba5cccbe915b7aece2a0ea5c5ff. Each frozen protocol pins that implementation commit, the actual Python3.12.14 executable hash and its analysis source. Canonical protocol hashes and freeze commands are in archive/protocol-freeze-01/result.json. Source readiness is not mathematical correctness or a completed fixed-case solve.

What was checked

Seven draft protocols passed the repository checker before freezing. The 35 Python sources in the seven prepared packages and two partial F1 sources were parsed with AST without importing their mathematical programs. Seven negative software fixtures confirm that absent/mismatched releases and changed protocol bytes fail before invoking a checkpoint process. Those are software checks, not real S3 gate passes. All eight preparation PDFs, nine pages total, were rendered with text-preserving compaction and visually inspected.

The new static mathematics is retained in theory-before-data.md and each track's derivation/source notes. In particular, one-anchor pole-three sampling has an exact rational-slope law and structured path incidence; it does not justify a random-graph collision model. The cover's infinity-difference class has zero norm but is nonzero, so matching norms cannot certify class equality. The lattice's two-summand capacity scale is distinguished from the earlier three-summand bound. These are scoped derivations, not new measured rates.

What blocks execution

The current repository rule requires an actual S3 resume checkpoint before runs. Automatic approval review rejected the attempted upload because specific disclosure consent was required. The explicit owner question remains pending; no upload occurred and no skip path was used. Read-only configuration checks also found no supported checkpoint upload token or AWS CLI/profile/key pair. Approval and a configured transport are both necessary before the real gate.

F1 has an additional platform halt recorded in environment/agent-halts-01.json; the coordinator's later capacity error is operational too. The partial F1 source is preserved byte-for-byte and its previous depth-four outcomes remain unknown. No missing execution is represented as an empty solution set.

RESUME.md identifies exact source/protocol custody and the next gated action. Tracking PR: https://github.com/isogeny-labs/crypto-autoresearcher/pull/2561. No comparative performance result, end-to-end ECDLP advantage, formal proof certificate, canonical scientific review or official goal/hypothesis transition is claimed. Native-model bindings remain unprobed.